



SEMESTER END EXAMINATIONS – MARCH 2024

Program	: B.E. – Artificial Intelligence and Data Science / Artificial Intelligence and Machine learning	Semester	: III
Course Name	: Database Management Systems	Max. Marks	: 100
Course Code	: AD32 / AI32	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

1. a) List and explain main characteristics of database approach. CO1 (06)
 b) Draw an ER diagram by considering the following set of requirements for a BANK database: CO1 (08)
 - i) Banks are identified by a name, code and the address of the main office.
 - ii) Banks have customers identified by name, customer ID, phone number and address.
 - iii) Customer can have one or more accounts, which are identified by an account number, account type and a balance.
 - iv) Customer can avail loans which are identified by loan ID, loan type and an account.
 - v) Banks may have branches identified by a branch name, branch number and an address.
 - vi) Accounts and loans are related to the bank branches.
- c) Discuss the significance and representation of the following with an example: CO1 (06)
 - i) Recursive relationship type
 - ii) Composite attributes
 - iii) Super Key.
2. a) What is data independence? Explain the concept of data independence using three schema architecture CO1 (08)
 b) Draw ER diagram for the following application: CO1 (06)
 Puppy pet shop wants to keep track of dogs and their owners. The person can buy any number of pet dogs. We store person's name, SSN and address and dog's name and sex. The owner of the pet dogs will be identified by SSN since the dog's names are not distinct.
 c) With the help of examples, explain different degrees in relationship type. CO1 (06)

UNIT - II

3. a) Consider the following relational database schema consisting of the three relation schemas CO2 (08)
 Suppliers(sID, sName, address)
 Parts(pID, pName, colour)
 Catalog(sID, pID, price)
 Answer the following questions using relational algebra queries:
 - i) Find the names of all red parts.
 - ii) Find the names of all suppliers who supply a part that is red or green.
 - iii) Find the sIDs of all suppliers who supply a part that is red or green.
- b) Briefly discuss the characteristics of a relational model. CO2 (06)
- c) Explain the significance of aggregate function operation in relational algebra with an example. CO2 (06)

4. a) Consider the following relational database schema: CO2 (08)
 Sailors (sid: integer, sname: string, rating: integer, age: real)
 Boats (bid: integer, bname: string, color: string)
 Reserves (sid: integer, bid: integer, day: date)
 Answer the following questions using relational algebra queries
 i) Find the names of sailors who have reserved boat 103.
 ii) Find the names of sailors who have reserved a red or a green boat.
 iii) Find the sids of sailors with age over 20 who have not reserved a red boat.
 b) Illustrate how constraints are violated while performing Insert and delete operations on relational model. CO2 (06)
 c) Justify the statement "UNION, INTERSECTION, and DIFFERENCE operations require that the relations on which they are applied be union compatible". CO2 (06)

UNIT - III

5. a) Explain the 6 clauses of an SQL retrieval query. Discuss which of the clauses are required and which are optional. CO3 (06)
 b) Consider the following tables: CO3 (08)
 BRANCH (branch-name:string, branch-city:string, assets:real)
 ACCOUNT (accno:int, branch-name:string, balance:real)
 DEPOSITOR (customer-name:string, accno:int)
 CUSTOMER(customer-name:string, customer-street:string, customer-city:string)
 LOAN (loannumber:int, branch-name:string, amount:real)
 BORROWER (customer-name:string, loan-number:int)
 Write the SQL Queries for the following:
 i) List in alphabetical order all customers who have a loan at 'Perryridge' branch.
 ii) Find the names of all customers who street address includes the substring "Main".
 iii) List the loan numbers in descending order of amount.
 iv) Find the average account balance at each branch.
 c) Explain different set operations in SQL with an example for each. CO3 (06)
6. a) STUDENT (regno: string, name: string, major: string, bdate:date) CO3 (12)
 COURSE (course#:int, cname: string, dept: string)
 ENROLL (regno: string, course#:int, sem:int, marks:int)
 BOOK _ ADOPTION (course#: int, sem:int, book-ISBN:int)
 TEXT (book-ISBN:int, book-title: string, publisher: string, author: string)
 Write SQL queries for the following:
 i) List the students name who have enrolled for 2nd sem.
 ii) List no of books adopted in semester '4'.
 iii) List no of books adopted by CSE department.
 iv) List the maximum no of courier adopted by any student.
 v) List the courses where course name begins with 'a' being held by 'cs' department.
 vi) List department which have more than 2 courses.
 b) Illustrate the following with syntax and examples CO3 (08)
 i) On delete Set NULL and On Update Cascade
 ii) Group by and Having

UNIT- IV

7. a) Determine the normal form for the following relation and justify the same CO4 (08)

ISBN $\rightarrow$ Title
 ISBN $\rightarrow$ Publisher
 Publisher $\rightarrow$ Address

BOOK

ISBN	Title	Publisher	Address
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- b) Briefly discuss different Informal Design Guidelines for Relational Databases. CO4 (06)
- c) Given a relation R(A, B, C, D) and Functional Dependency set FD = { AB $\rightarrow$ CD, B $\rightarrow$ C }, determine whether the given R is in 2NF? If not convert it into 2 NF. CO4 (06)

8. a) Consider the following relation for published books: CO4 (08)

BOOK (Book_title, Authorname, Book_type, Listprice, Author_affil, Publisher)

Author_affil refers to the affiliation of the author.

Suppose the following dependencies exist:

Book_title $\rightarrow$ Publisher, Book_type

Book_type $\rightarrow$ Listprice

Author_name $\rightarrow$ Author-affil

What normal form is the given relation in? Apply normalization until you cannot decompose the relations further. State the reasons behind each decomposition.

- b) Briefly discuss Armstrong's Inference Rules for Functional Dependency. CO4 (06)
- c) Explain the general guidelines for a relation in second and third normal form with an example for each. CO4 (06)

UNIT - V

9. a) Consider the three transactions T1, T2, and T3, and the schedules S1 and S2 given below. Draw the serializability (precedence) graphs for S1 and S2, and state whether each schedule is serializable or not. If a schedule is serializable, write down the equivalent serial schedule(s). CO5 (06)

T1: r1 (X); r1 (Z); w1 (X);

T2: r2 (Z); r2 (Y); w2 (Z); w2 (Y);

T3: r3 (X); r3 (Y); w3 (Y);

S1: r1 (X); r2 (Z); r1 (Z); r3 (X); r3 (Y); w1 (X); w3 (Y); r2 (Y); w2 (Z); w2 (Y);

S2: r1 (X); r2 (Z); r3 (X); r1 (Z); r2 (Y); r3 (Y); w1 (X); w2 (Z); w3 (Y); w2 (Y);

- b) Discuss deadlock prevention protocols. CO5 (07)
- c) Explain with a state transition diagram of the transaction and its execution states. CO5 (07)

10. a) Which of the following schedules is (conflict) serializable? For each serializable

Schedule, determine the equivalent serial schedules.

i) r1(X); r3(X); w1(X); r2(X); w3(X);

ii) r1(X); r3(X); w3(X); w1(X); r2(X);

iii) r3(X); r2(X); w3(X); r1(X); w1(X);

iv) r3(X); r2(X); r1(X); w3(X); w1(X);

- b) Define transaction. Why concurrency control is required explain with a suitable example. CO5 (07)
- c) Explain the process of starvation and detection techniques. CO5 (07)
